ASTRONOMY OLYMPIAD 2026

SENIOR CATEGORY — PRACTICE PAPER

Unofficial preparation material • Created for workshop practice, not an official competition paper

Suggested duration: 35 minutes **Total:** 30 marks

Instructions
• Answer all questions.
• Section A: 15 multiple-choice questions, 1 mark each (15 marks).
• Section B: 5 short-answer questions, 3 marks each (15 marks).
• Show working for numerical questions. Award partial credit for correct reasoning.
• For this practice paper, there is no negative marking.

Section A — Multiple Choice (15 marks)

1. A planet completes one orbit in 2 years. Its average orbital distance is approximately how many AU? Use $T^2 = a^3$, with T in years and a in AU.
A. 1.26 AU B. 1.59 AU C. 2 AU D. 4 AU

2. What keeps planets in orbit around the Sun?
A. Magnetism alone B. Gravity combined with their forward motion C. Air resistance D. The Sun's light pressure alone

3. If a star's surface temperature increases, its peak emission shifts toward which part of the spectrum?
A. Longer wavelengths B. Shorter wavelengths C. No change D. Only radio waves

4. A star has parallax 0.5 arcseconds. Its distance is…
A. 0.5 parsec B. 2 parsecs C. 5 parsecs D. 10 parsecs

5. What does a star's apparent brightness depend on most directly?
A. Its distance and luminosity B. Its name C. Only its colour D. Only its age

6. Which process powers the Sun's core?
A. Chemical burning B. Nuclear fission C. Hydrogen fusion into helium D. Radioactive decay only

7. Why does a massive star generally have a shorter lifetime than a low-mass star?
A. It contains no hydrogen B. It uses its fuel much faster C. It is always colder D. It has no gravity

8. A galaxy's spectrum is shifted toward longer wavelengths. This is commonly called…
A. blueshift B. redshift C. parallax D. eclipse

9. What is the main evidence that the universe is expanding?
A. All stars are identical B. Many distant galaxies show redshift related to distance C. The Moon has craters D. Saturn has rings

10. Which method can detect an exoplanet by measuring a small, repeated dip in a star's brightness?
A. Transit B. Direct radio broadcast C. Lunar eclipse D. Parallax alone

11. Why are radio telescopes useful?
A. They detect radio wavelengths from space B. They work only underwater C. They eliminate gravity D. They can see only the Moon

12. What is a black hole's event horizon?
A. A solid surface B. The boundary beyond which light cannot escape to a distant observer C. The centre of every galaxy D. A cloud of ice

13. The Sun's energy output is called its…
A. parallax B. luminosity C. declination D. eccentricity

14. What is dark matter inferred to do on galactic scales?
A. Emit bright visible light B. Contribute gravity without emitting detectable light in the usual way C. Stop galaxies

rotating D. Destroy all stars

15. What is the main reason seasons differ between hemispheres?
A. Earth’s tilt makes the hemispheres receive different sunlight angles and day lengths B. The Sun turns off in winter C. Earth’s orbit stops D. The Moon changes Earth’s temperature directly

Section B — Short Answer & Simple Problems (15 marks)

16. State Kepler’s third law in its simple Solar System form.

Answer: _____

_____

17. A star has luminosity 4 times the Sun’s and is 2 times as far away as the Sun would be from an observer. Relative to the Sun at its reference distance, compare its observed flux using $F \propto L/d^2$.

Answer: _____

_____

18. Why can infrared telescopes reveal objects hidden by some clouds of dust?

Answer: _____

_____

19. Give one difference between a white dwarf and a neutron star.

Answer: _____

_____

20. A galaxy is observed at redshift z = 0.1. Using the simple low-redshift approximation $v \approx cz$ and $c \approx 300{,}000$ km/s, estimate its recession speed.

Answer: _____

_____

SENIOR CATEGORY — ANSWER KEY & EXPLANATIONS

Use this section after students have completed the paper.

Section A — Multiple Choice

1. B. $a^3 = 4$, so a = 4^(1/3) ≈ 1.59 AU.

2. B. Gravity continually changes the direction of a planet's motion.

3. B. Wien's law: hotter objects peak at shorter wavelengths.

4. B. Distance in parsecs = 1 / parallax in arcseconds.

5. A. Observed brightness depends on luminosity and distance.

6. C. The Sun is powered mainly by nuclear fusion.

7. B. High-mass stars have much higher luminosities and consume fuel rapidly.

8. B. Redshift means spectral features are observed at longer wavelengths.

9. B. The observed relation between distance and recession speed supports expansion.

10. A. A planet passing in front of its star blocks a small fraction of its light.

11. A. Radio telescopes study radio emission from astronomical sources.

12. B. It marks the boundary of no return.

13. B. Luminosity is the total energy emitted per unit time.

14. B. Its gravitational effects help explain observations such as galaxy rotation curves.

15. A. Axial tilt changes seasonal insolation.

Section B — Suggested Answers

16. $T^2 = a^3$ when T is measured in years and a in astronomical units.

17. Relative flux = $4 / 2^2 = 1$; it would have the same observed flux.

18. Infrared wavelengths can pass through some dust more effectively than visible light and can detect cooler objects.

19. A white dwarf is the compact remnant of a low/intermediate-mass star; a neutron star is a much denser remnant usually formed after a massive star's core collapse.

20. v ≈ 0.1 × 300,000 = 30,000 km/s. This is a simplified approximation.

Marking guidance: Each short-answer question is worth up to 3 marks. Give credit for scientifically correct equivalent wording. For numerical questions, award partial credit for a correct formula or method even if the final arithmetic is imperfect.